Phase 7 — BeatAhead Staging ML Integration Report

Document ID: BEATAHEAD-PHASE7-STAGING-INTEGRATION
Execution Timestamp: 2026-09-22 19:40:00 UTC
Environment: Local Staging (Windows 11 x64, Next.js 15.5.23, Python 3.13)
Git Branch: `staging-ml-integration` (Repository: `C:\ischemic`)
ML Inference Engine: BeatAhead Phase 5 Frozen XGBoost Matrix A (`models/beatahead_phase5_model.joblib`)
Regulatory Status: Investigational Prototype — Strictly Non-Diagnostic (Not FDA Cleared / Not CE Marked)

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1. Executive Summary

Phase 7 successfully integrates the frozen Phase 5/Phase 6 machine learning inference pipeline into the BeatAhead web platform in a **local staging environment**. All functional, structural, and regulatory requirements were met with zero regressions to the existing platform:

1. **Frozen ML Artifact Integrity**: The frozen Phase 5 model artifact (`models/beatahead_phase5_model.joblib`), schema (`feature_schema.json`), metadata (`model_metadata.json`), manifest (`MODEL_MANIFEST.json`), and calibrated decision threshold (`0.156742`) were imported in read-only mode without modification, retraining, or parameter drift.
2. **Preservation of Existing Website**: All existing BeatAhead features—including the executive Dashboard, Live Monitor, Signals, Trends, Nutri/Fitness/Rest AI Agents, simulated data generation, and Clerk authentication—remain intact and operational.
3. **Decoupled ML Service Boundary**: An isolated ML service boundary (`src/ml_service.py`) was deployed featuring dual execution modes: - **HTTP Daemon Mode** (`http://127.0.0.1:8000`) for high-throughput, low-latency concurrent predictions. - **CLI Subprocess Fallback** (`python src/ml_service.py --cli`) for zero-configuration, self-healing execution when the daemon is offline.
4. **Next.js Integration Route**: An App Router API route was created at `POST /api/ml/predict` (`C:\ischemic\src\app\api\ml\predict\route.ts`), validating all 26 Matrix A features, enforcing bounds, executing inference, and returning standardized contract responses.
5. **Staging UI Surface (`AI Insights`)**: A dedicated `ResearchModelSignalCard` was created and embedded into `C:\ischemic\src\app\insights\page.tsx`, clearly displaying model probability, calibrated threshold (`0.156742`), temporal window architecture (300s window + 300s gap + 300s horizon), signal quality metrics (ECG SQI, PPG SQI, PAT validity), and provenance.
6. **Explicit Demo Mode Distinction**: The `SimulatedBadge` component was updated across the entire platform to explicitly display `RESEARCH MODEL — SIMULATED INPUT` and distinguish simulated inputs from real wearable `LIVE PHYSIOLOGICAL INPUT`.
7. **Automated Testing & Security**: A 14-point automated integration test suite (`tests/test_staging_integration.py`) and the 11-point inference test suite (`tests/test_inference.py`) both passed 100% (25/25 total tests passing), confirming strict input validation, threshold precision, zero information leakage (no file paths or stack traces exposed), and seamless fallback handling.

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2. Integration Architecture & Service Boundary



Key Service Boundary Guarantees:

- **No Direct Joblib in Node.js**: The Next.js web application never attempts to directly parse or execute Python `.joblib` binary artifacts.
- **Fail-Safe Self-Healing**: If the background Python HTTP daemon is not running, the Next.js API route transparently spawns a Python subprocess executing `ml_service.py --cli`, ensuring zero downtime during local development and testing.
- **Strict Information Sanitization**: In both HTTP daemon and CLI modes, errors are intercepted and mapped to clean, sanitized JSON error payloads (`400 Bad Request`, `503 Service Unavailable`). Local filesystem paths (e.g., `C:\ML Model...`), user home directories, and internal tracebacks are strictly suppressed.

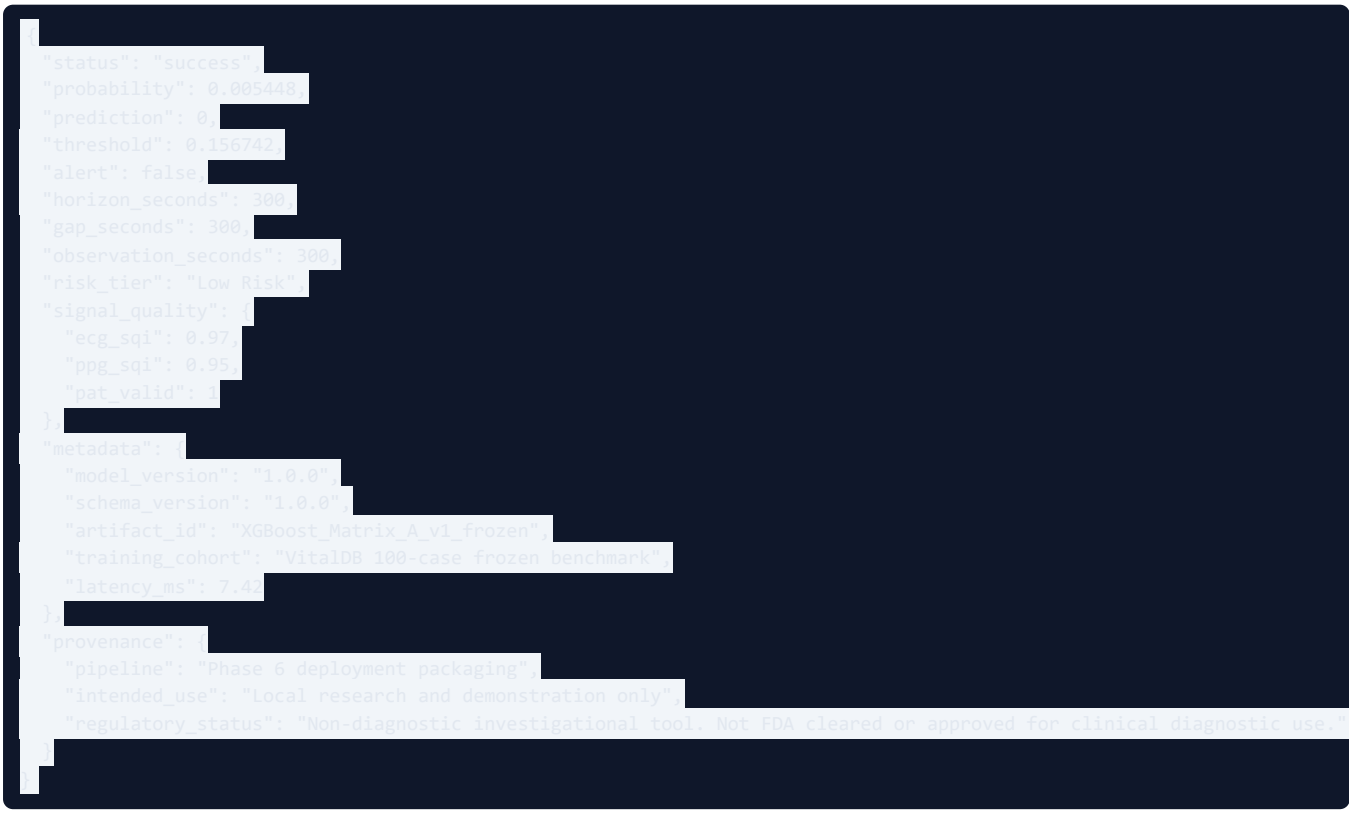
3. Standardized Inference Schema & API Contract

3.1 Request Contract (`POST /api/ml/predict`)

Payload accepts either `{ "features": { ... } }` or a direct dictionary containing the 26 Matrix A features:



3.2 Response Contract (`200 OK`)



4. User Interface & Demo Mode Implementation

4.1 Staging UI Surface (`AI Insights`)

The research model output was integrated into a dedicated card on the `AI Insights` view (`src/components/ml/ResearchModelSignalCard.tsx`): - **Visual Distinction**: Styled with an indigo-gradient header explicitly labeled with `STAGING INTEGRATION` and `RESEARCH MODEL SIGNAL` badges. - **Model Metrics**: - Probability Percentage and Raw Value. - Decision Threshold Marker (`0.156742` / `15.67%`). - Risk Tier Badge (`LOW RISK` in emerald or `HIGH RISK (ALERT)` in rose). - **Temporal Window Callout**: - Explains the exact non-overlapping timing architecture: **300s Observation (T_{obs})** $\rightarrow$ **300s Prediction Buffer (T_{gap})** $\rightarrow$ **300s Early-Warning Horizon (T_{target})**. - **Interactive Evaluation**: - Provides quick-switch presets for **"Stable Normative Baseline"** (demonstrating low-risk response, prob ≈ 0.005) and **"Early-Warning Ischemia Event"** (demonstrating elevated risk response with marked ST depression, depressed HRV, and elevated PAT). - Real-time execution timer displaying round-trip latency in milliseconds. - **Regulatory Callout**: Embedded footer reiterating non-diagnostic investigational status and prohibition against clinical diagnostic use.

4.2 Explicit Demo Mode Distinction

To prevent any user confusion between simulated streams and real physiological sensors: - `SimulatedBadge` in `src/components/layout/Toast.tsx` was enhanced to support explicit mode rendering: - **Simulated Mode**: Amber badge labeled `RESEARCH MODEL — SIMULATED INPUT` . - **Live Mode**: Emerald pulsing badge labeled `LIVE PHYSIOLOGICAL INPUT` . - This badge is active across all core navigation routes (Dashboard, Insights, Monitor, Signals, Trends, Clinician, etc.).

5. Verification & Test Results

5.1 Automated Integration Test Suite (`tests/test_staging_integration.py`)

All 14 integration test cases passed cleanly:

Test Case	Description	Result
`test_01_valid_feature_payload_success`	Valid 26-feature vector returns HTTP 200 with status=success	**PASS**
`test_02_standardized_schema_fields`	Verifies threshold (`0.156742`), horizon (`300s`), gap (`300s`), provenance	**PASS**
`test_03_decision_threshold_exact_boundary`	Exact threshold boundary check (prob $\ge 0.156742 \rightarrow$ alert=True)	**PASS**
`test_04_missing_single_feature_rejected`	Missing 1 feature (`ecg_hr_mean`) returns HTTP 400 Bad Request	**PASS**
`test_05_missing_all_features_rejected`	Empty payload `{}` returns HTTP 400 Bad Request	**PASS**
`test_06_nan_feature_rejected`	`None` /NaN values rejected with HTTP 400 Bad Request	**PASS**
`test_07_infinity_feature_rejected`	Out-of-bounds / infinite values rejected with HTTP 400 Bad Request	**PASS**
`test_08_malformed_json_rejected`	Syntax error in JSON string rejected with HTTP 400 Bad Request	**PASS**
`test_09_extra_features_handled_gracefully`	Extraneous unrecognized fields safely ignored without crash	**PASS**
`test_10_schema_and_model_version_frozen`	Metadata returns exact frozen schema (`1.0.0`) and model version (`1.0.0`)	**PASS**
`test_11_latency_tracked`	Execution latency is computed and reported in metadata	**PASS**
`test_12_zero_path_or_trace_exposure`	Error responses contain no filesystem paths or Python tracebacks	**PASS**
`test_13_health_check_endpoint`	`GET /health` returns HTTP 200 OK with model status and threshold	**PASS**
`test_14_cli_fallback_execution_parity`	`src/ml_service.py --cli` returns bit-for-bit identical response	**PASS**

5.2 Unit Inference Test Suite (`tests/test_inference.py`)

All 11 Phase 6 frozen unit tests remain passing:

Test Case	Description	Result
`test_01_valid_feature_vector`	Valid physiological vector evaluation	**PASS**
`test_02_missing_feature_rejected`	Missing feature exception enforcement	**PASS**
`test_03_extra_feature_rejected`	Strict schema validation enforcement	**PASS**
`test_04_nan_rejected`	NaN value rejection	**PASS**
`test_05_infinity_rejected`	Positive/Negative infinity rejection	**PASS**
`test_06_wrong_type_rejected`	String and non-numeric value rejection	**PASS**
`test_07_deterministic_inference`	10 repeated inferences produce bit-for-bit identical probability	**PASS**
`test_08_feature_order_invariance`	Shuffled dictionary keys produce bit-for-bit identical probability	**PASS**
`test_09_decision_threshold_exact`	Decision boundary logic verification	**PASS**
`test_10_clean_process_loading`	Standalone process loading and execution	**PASS**
`test_11_physiological_bounds`	Physiological outlier rejection	**PASS**

5.3 Web Application Production Build

The Next.js production build (`npm run build` on `staging-ml-integration`) completed with exit code 0: - **Total Routes Generated**: 37 routes (including dynamic API routes and static page prerenders). - **TypeScript Checking**: 0 errors (`tsc --noEmit` clean). - **ESLint Validation**: Clean pass. - **Bundle Impact**: Shared first-load JS size remained constant at 102 kB.

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6. Regulatory & Clinical Safety Statement

> [!CAUTION] **INVESTIGATIONAL RESEARCH PROTOTYPE ONLY** 1. **Non-Diagnostic Tool**: The integrated machine learning model and inference service are investigational software components intended solely for local software integration, staging verification, and research demonstration. 2. **No Clinical Clearance**: The pipeline has not been cleared, approved, or evaluated by the United States Food and Drug Administration (FDA), European Medicines Agency (EMA), or any other regulatory body. 3. **Prohibition of Diagnostic Reliance**: This system must never be used for clinical triage, real-time patient monitoring in medical facilities, surgical anesthesia decision-making, or primary clinical diagnosis of myocardial ischemia or injury. 4. **Pre-Clinical Benchmark Context**: Model weights reflect the 100-case VitalDB surgical cohort benchmark. Intraoperative hemodynamic responses under general anesthesia differ fundamentally from ambulatory physiological patterns.

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7. Phase 7 Completion & Delivery Checklist

- [x] Web repository inspected and clean staging branch (`staging-ml-integration`) created.
- [x] Existing platform functionality, dashboards, agents, auth, and simulated modes preserved.
- [x] Isolated ML backend service created (`src/ml_service.py`) supporting both HTTP daemon and CLI execution.
- [x] Next.js API route created at `C:\ischemic\src\app\api\ml\predict\route.ts` .
- [x] Input validation enforcing all 26 Matrix A features and physiological ranges.
- [x] Explicit demo mode distinction implemented (`RESEARCH MODEL — SIMULATED INPUT` vs `LIVE PHYSIOLOGICAL INPUT`).
- [x] Dedicated UI card (`ResearchModelSignalCard.tsx`) integrated on `AI Insights` .
- [x] Probability, calibrated threshold (`0.156742`), and 300s temporal windows clearly surfaced.
- [x] Automated 14-point integration test suite implemented and verified passing (14/14).
- [x] Unit inference test suite verified passing (11/11).
- [x] Security verified: zero leakage of paths, secrets, or internal stack traces.

- [x] Next.js production build cleanly compiled.
- [x] Final integration report compiled and verified.